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Bachelor Thesis

A Structural Dynamic Model of Job-Search Behaviour Among Unemployed Graduates

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Abstract

This thesis examines job-search behaviour among newly graduates receiving unemployment insurance (UI) in Denmark using a dynamic structural job-search model estimated on Danish administrative register data. The model describes how forward-looking unemployed graduates choose job-search intensity in response to current and expected future income while unemployed. It assumes hand-to-mouth consumption, fixed wage, and incorporates unobserved heterogeneity through search costs. Using variation generated by the 2023 reform of the Danish graduate UI scheme, the model is jointly estimated on empirical hazards from the pre- and post-reform regimes to identify behavioural responses to differences in benefit levels and potential benefit duration.

Weekly empirical hazards of transition from unemployment to employment are estimated for three groups of newly graduates: non-parents below age 30, non-parents aged 30 or above, and parents. The estimated hazards exhibit clear duration dependence and pronounced responses around benefit exhaustion. The structural parameters are estimated using the Generalized Method of Moments estimator by matching model-implied hazards to the empirical hazards.

The benchmark model with log utility reproduces the broad decline in employment-exit hazards and the increase in search intensity near benefit exhaustion, although it understates the magnitude of the exhaustion spike. The estimated model also generates policy responses that are broadly consistent with existing reduced-form evidence. The estimated model is used to examine how counterfactual changes in benefit levels and potential benefit duration affect search intensity and expected unemployment duration. Overall, the thesis shows that a relatively simple dynamic structural model can explain important features of job-search behaviour among unemployed graduates in Denmark, while also identifying aspects of the observed hazard profile that require further extensions of the model.

1 Introduction

The design of unemployment insurance (UI) for newly graduated individuals raises an important policy question: how do benefit levels and benefit duration affect the transition from unemployment to employment? Graduates may need income support if they do not enter employment immediately after completing their education. At the same time, the level and potential duration of UI benefits may affect their incentives to search for and enter employment. The recent reforms of the Danish graduate UI scheme in 2017 and 2023 reflect an ongoing policy interest in reducing unemployment duration and increasing labour supply among graduates. Understanding how their job-search behaviour responds to different benefit schedules, and developing a framework for analysing alternative UI designs, is therefore of direct policy relevance. To study this question, this thesis adapts a dynamic structural job-search model to the Danish graduate UI scheme and estimates it using Danish administrative register data on newly graduated UI recipients.

This thesis builds on a broader literature that studies unemployment dynamics through job-search behaviour. Dynamic structural job-search models provide a way to connect observed hazards to underlying search incentives and to examine how alternative UI policies affect job-search behaviour. Lentz (2009) estimates a structural search model using Danish register data and relates observed unemployment spells to model-implied search effort in order to recover the underlying search behaviour and structural parameters of Danish UI recipients. His model is richer than the one used in this thesis, as it includes both search effort and savings as choice variables. However, Lentz’s analysis is based on older register data from 1981–1994 and studies a UI regime with substantially longer potential benefit durations than the current graduate UI system. In a more recent paper, DellaVigna et al. (2017) develop a dynamic structural job-search model using data and reform variation from the Hungarian UI system. Their model accounts for declining unemployment hazards and responses around changes in the benefit schedule.

More recent Danish studies examine specific dimensions of job-search behaviour among UI recipients. Using data on actual job applications, Fluchtmann et al. (2024) show that applied-for wages and job characteristics vary with unemployment duration and highlight the importance of worker heterogeneity and dynamic selection. Rosholm and Svarer (2008) show that active labour-market policies also affect job-finding behaviour. These studies provide important evidence on application choices and policy responses. However, there is less recent evidence using structural models to explain the dynamics of employment exits over the unemployment spell or responses to alternative benefit schedules in the Danish UI system, particularly among newly graduated UI recipients.

Most of the structural job-search literature studies general UI recipients rather than specific subgroups within the UI system. Newly graduated UI recipients are a relevant case in Denmark as they face a separate benefit schedule, and because recent Danish reforms have specifically targeted this group. Existing evaluations of graduate UI reforms, such as STAR (2022) and Sørensen (2025), provide reduced-form evidence on the effects of implemented reforms. However, they do not by themselves provide a structural framework for linking the observed responses to underlying job-search incentives or for examining counterfactual graduate UI schedules.

The contribution of this thesis is to bring a structural search framework to the specific setting of unemployed graduates in the Danish graduate UI scheme, using recent register data and variation from the 2023 reform. The model provides a first step towards explaining the employment-exit hazard profile of newly graduated UI recipients through a dynamic framework and can be used to analyse model-implied responses to counterfactual changes in benefit duration and benefit levels.

Using Danish administrative register data, I first estimate empirical hazards of exit from unemployment to wage employment for newly graduated UI recipients before and after the 2023 reform. The hazards show two pronounced patterns. After an initial increase during the first weeks of unemployment, they decline over much of the spell, suggesting the presence of dynamic selection. They also rise sharply around UI exhaustion in both reform regimes, indicating that the timing of exhaustion is important for job-search behaviour.

These patterns motivate a forward-looking dynamic job-search model in which job-seekers choose search intensity in response to the current and future benefit schedule. The model follows DellaVigna et al. (2017) and includes unobserved heterogeneity in search costs to generate declining aggregate hazards through dynamic selection. Since the specification presented in this thesis cannot reproduce the sharp initial increase in employment exits, I treat the first weeks as a transition phase and estimate the model from week 7 and onward. The model is estimated separately for non-parents below age 30, non-parents aged 30 or above, and parents. The preferred log-utility specification captures the broad decline in hazards and the increase around UI exhaustion, but understates the exhaustion spike, especially for non-parents. A shifted-log specification improves the fit around exhaustion for the post-reform hazards, but produces less interpretable parameters with greater uncertainty.

The model-implied policy responses are broadly consistent with existing reduced-form evidence. For the 2017 benefit reduction, the benchmark model implies an elasticity of unemployment duration with respect to the benefit level of 0.43, compared with 0.44 in STAR (2022). The comparison with Sørensen (2025) also suggests that the model captures the general direction and magnitude of the 2023 reform effects. Because these reduced-form estimates are based on different empirical designs and are generally imprecisely estimated, the comparisons are best viewed as plausibility checks rather than formal out-of-sample validation.

Overall, the results suggest that a relatively simple dynamic search model can explain important features of job-search behaviour among newly graduated UI recipients, particularly the declining hazard after the initial transition phase and the timing of response around UI exhaustion. At the same time, additional mechanisms may be needed to explain the early transition into employment and to more accurately capture the magnitude of the exhaustion spike.

The remainder of the thesis is structured as follows. Section 2 presents the institutional background, data, sample construction, and empirical hazards. Section 3 sets up the dynamic structural job-search model, and Section 4 describes its estimation and identification. Section 5 presents the model fit and parameter estimates, compares the model-implied policy responses with existing reform evidence, and examines counterfactual graduate UI schedules. Section 6 discusses the model's limitations and directions for further research, while Section 7 concludes.

2 Institutional Background, Data, and Empirical Hazards

This section presents the institutional and empirical background for the analysis. I first describe the Danish graduate UI scheme and the benefit schedules faced by newly graduated UI recipients before and after the 2023 reform. I then describe the register data, sample construction, and definitions of unemployment spells and employment exits. The section concludes by presenting the empirical hazards of exit to employment, which provide the main empirical motivation for the structural model.

2.1 The Danish Graduate UI Scheme

In Denmark, newly graduated individuals can receive unemployment insurance benefits (*dagpenge*) through a special graduate scheme after completing their education. Under this scheme, they do not need to meet the usual income and employment requirements for eligibility under the regular UI scheme. Unlike the regular UI scheme, new graduates' benefit level does not depend on previous earnings. Instead, benefit levels are defined as a fixed share of the maximum UI benefit rate. Newly graduated individuals can become eligible for UI through the graduate scheme if they have completed a qualifying education with a prescribed duration of at least 18 months. In addition, individuals must be members of an unemployment insurance fund (*a-kasse*) and register as unemployed after graduation (A-kasser.dk, 2023).

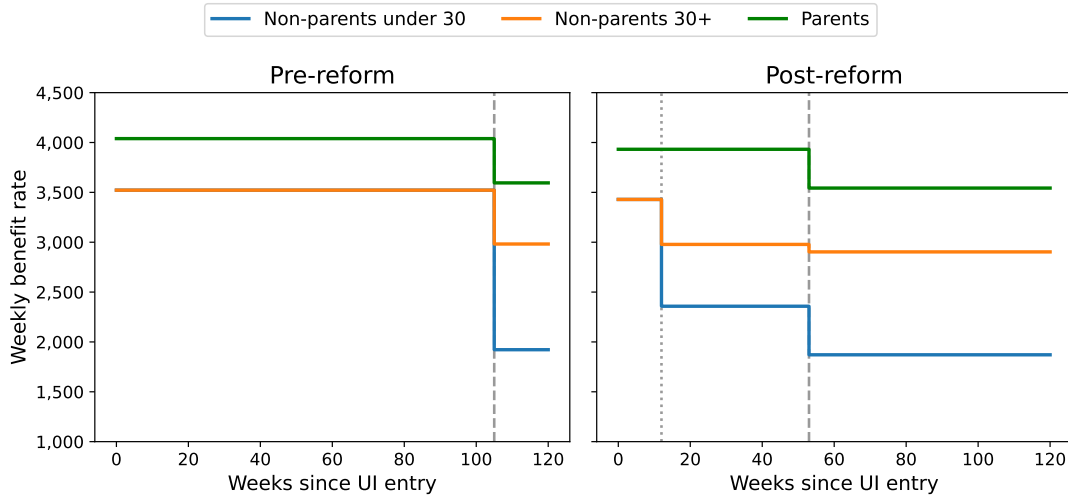
Over the last decade, there have been several reforms of the UI system that targets graduates in an attempt to decrease the length of unemployment and increase labour supply for this group. Before 2017, the UI graduate scheme was relatively simple: all graduates received the same benefit rate, corresponding to 82 percent of the maximum UI benefit, and the potential benefit duration (PBD) followed the general UI system. Since 2010, this duration has been two years. The first reform targeted at graduate UI was introduced in 2017. This reform created a distinction between parents and non-parents. Parents continued to receive 82 percent of the maximum UI benefit rate, while the benefit rate for non-parents was reduced to 71.5 percent of the maximum rate. The PBD remained two years. (STAR, 2022)

The most recent reform of the graduate UI system came into force in May 2023. The reform changed both PBD and benefit levels. First, the PBD for all newly graduated UI recipients was reduced from two years to one year. Second, the reform introduced a front-loaded benefit schedule for non-parents. Non-parents initially receive the same rate as before the reform, corresponding to 71.5 percent of the maximum UI benefit, but after three months the benefit level is reduced. The reform also introduced a distinction between non-parents below age 30 and non-parents aged 30 or above. For non-parents below age 30, the benefit rate is reduced by 31.1 percent after three months, while the corresponding reduction for non-parents aged 30 or above is 13.1 percent. (Akademikerne et al., 2023; A-kasser.dk, 2026)

2.2 Benefit Schedules and Reform Variation

This thesis focuses on the benefit regimes in place from January 2017 to May 2023, referred to as the pre-reform regime, and from May 2023 onward, referred to as the post-reform regime. Within these regimes, there are three distinct groups that face different benefit schedules: parents, non-parents below age 30, and non-parents aged 30 or above. Figure 1 illustrates the resulting benefit schedules for the three groups used in the analysis. Since social benefit rates are regulated each year, the benefit levels in Figure 1 are measured as weighted averages of yearly benefit rates converted to 2025 prices.

Figure 1: Graduate UI benefit schedules before and after the 2023 reform



Notes: The figure shows representative graduate UI benefit schedules before and after the 2023 reform. Benefit levels are expressed in 2025 prices and averaged across UI-entry years using their shares in the estimation sample as weights. The dotted line marks the week-12 benefit reduction, and the dashed lines mark UI exhaustion.

After UI exhaustion, individuals may receive social assistance (*kontanthjælp*) if they satisfy the relevant eligibility requirements. Eligibility depends, among other things, on household income, partner income, and wealth. The level of social assistance also varies by parental status and age. This is reflected in the post-UI-exhaustion benefit level, which differs across the three groups in both the pre- and post-reform regimes.

The variation across reform regimes provides policy variation that can help identify policy-invariant features of graduates job-search behaviour. In the empirical analysis, I split the estimation sample into pre-reform and post-reform periods to make use of this variation.

2.3 Register Data and Estimation Sample

The analysis is based on Danish administrative register data from Statistics Denmark and the Ministry of Employment. The main register used to define unemployment spells is DREAM (*Beskæftigelsesministeriets forløbsdatabase*), which contains weekly information on public transfer and labour market status for the Danish population. The education register, KOTRE (*Komprimeret elevregister*) is used to identify newly graduated individuals who have completed a qualifying education. Wage information is obtained from the BFL register (*Detaljeret lønmod-*

tagerdata fra e-Indkomst), while demographic background variables are obtained from the BEF register (*Befolkningen*).

The starting point for the sample is individuals who complete a qualifying education between January 2017 and December 2025 and subsequently enter UI. DREAM data were available up to January 31, 2026, which defines the end of the observation window. Individuals below age 18 and individuals not observed in the Danish population in the year of UI entry are excluded. Graduate UI entry is defined as the first DREAM week where UI is received within a window from 10 weeks before to 26 weeks after the registered graduation date. This window is used because the registered graduation date may differ from the actual completion date, and because some graduates may delay UI take-up after graduation. The estimation sample is divided into the three groups that face different benefit schedules: non-parents below age 30, non-parents aged 30 or above, and parents based on their age and parental status at the time of UI entry.

For the sample used in the estimation, I exclude periods affected by the COVID-19 pandemic. The final pre-reform estimation period consists of January 2017–February 2018 and September 2021–April 2023, while the post-reform period runs from May 2023 to December 2025. Motivation for the chosen excluded periods and effects on sample composition can be found in Appendix A. The final estimation sample contains 159,090 graduate UI entries, where 79,512 are observed in the pre-reform period and 79,578 in the post-reform period.

Table 1: Sample composition and baseline characteristics by group

		Pre-reform			Post-reform		
		Non-parents under 30	Non-parents 30+	Parents	Non-parents under 30	Non-parents 30+	Parents
Graduate UI entrants	<i>N</i>	54,249	9,479	15,784	56,947	9,397	13,234
Group share	%	68.2	11.9	19.9	71.6	11.8	16.6
Female	%	57.8	52.2	72.2	58.3	52.8	73.1
Mean age		25.7	35.2	33.7	25.9	34.8	34.1
SD		(2.0)	(7.4)	(6.5)	(2.0)	(7.4)	(6.5)
<i>Education</i>							
Vocational	%	14.0	17.4	21.7	11.9	18.7	23.3
Short higher	%	6.0	7.7	6.6	6.0	7.9	7.9
Medium higher	%	35.8	33.7	42.9	30.5	30.2	40.4
Long higher	%	44.3	41.2	28.7	51.6	43.2	28.4

Notes: Graduate UI entrants is all new graduates that enter UI after graduation in the given reform period. Education shares refer to the distribution of education within each group and period.

Table 1 reports the composition and baseline characteristics of the estimation sample. Non-parents below age 30 constitute the largest group in both reform regimes, accounting for 68.2 percent of the pre-reform sample and 71.6 percent of the post-reform sample. This group therefore receives substantial weight in the aggregate hazard when all three groups are pooled. The smaller sample sizes for non-parents aged 30 or above and parents limit how much additional heterogeneity can be introduced within these groups in the structural estimation. The composition of each group is broadly stable across the pre- and post-reform periods in terms of gender, age, and education. Only around 40% of all graduates enter UI after graduation. Since

the estimation sample only includes graduates who enter UI after graduation, it should not be interpreted as representative of all newly graduated individuals.

2.4 Defining Graduate Unemployment Spells and Employment Exits

Unemployment duration is measured in weeks from UI entry. A UI spell is treated as ongoing until the individual has six consecutive weeks without receiving UI, so short interruptions are not treated as exits from unemployment. Since the timing of UI exhaustion is central to the analysis, weeks without UI receipt are excluded from the duration count within otherwise continuous UI spells. The unemployment spell may continue after graduate UI exhaustion if the individual transitions to another job-search benefit, defined as a benefit where the recipient is unemployed and expected to actively search for work. If the individual enters such a benefit within 26 weeks after the UI spell ends, without having a stable employment spell or entering a non-job-search benefit during the gap, the unemployment spell is treated as continuous. Non-job-search benefits include social transfers where recipients are not expected to actively seek or take up employment, such as parental leave benefits, sickness benefits, educational support, and similar benefits. The unemployment spell ends once the individual has six consecutive weeks without receiving job-search benefits.

The outcome of interest is exit from unemployment to employment. An unemployment spell is classified as ending in employment if the individual is observed with at least three weeks of employment within ten weeks after the unemployment spell ends. Exits to non-job-search benefits and exits to self-sufficiency are treated as right-censored. Individuals who do not exit unemployment before the end of the observation window are also right-censored.

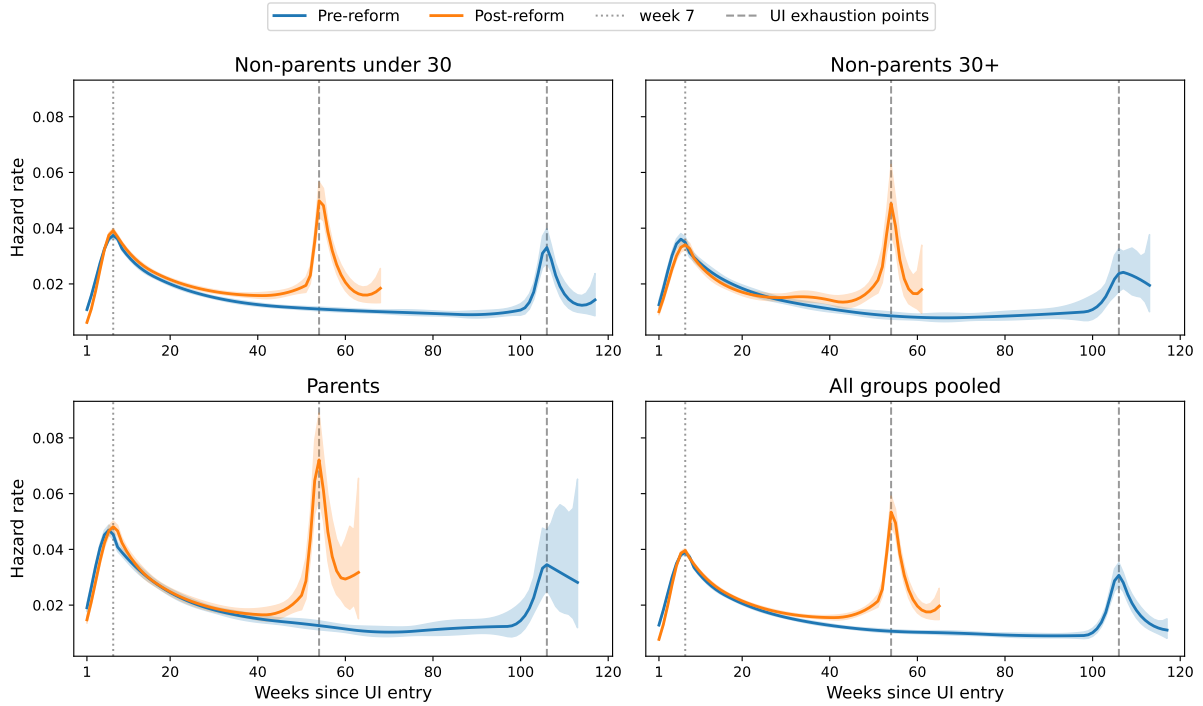
2.5 Empirical Hazards of Exit to Employment Among Graduates

Figure 2 presents the empirical weekly hazards of exit from unemployment to employment. I estimate these using a semi-parametric discrete-time logit model, with B-splines in unemployment duration to allow the hazard to vary flexibly over the spell (Berger & Schmid, 2017; Carter & Signorino, 2010). The technical details of this hazard estimation are presented in Appendix C.

The empirical hazards reveal several interesting patterns in job-search behaviour among the unemployed graduates. First, the exit probability is very low at the beginning of the unemployment spell but rises sharply over the following seven weeks. This initial increase appears across groups and reform regimes and is of a similar magnitude. To examine how persistent this pattern is, I examined the hazards separately by month of UI entry and education type. The size of the increase varied across these splits, but in all cases the hazard started at a very low level and rose to a peak around week 7. This suggests that the initial increase may reflect mechanical frictions between job search and job entry.

After this initial phase, the hazard generally declines over the unemployment spell until it approaches the time of exhaustion of UI benefits, suggesting that the graduates have more intensive job-search behaviour at the beginning of their UI duration which then decreases as time on UI goes by. Around the point of exhaustion the hazards increase sharply. In the pre-reform regime, this increase occurs around the end of the two-year benefit period, while in the post-reform regime it occurs around the end of the one-year benefit period. This pattern suggests

Figure 2: Empirical hazards of exit from unemployment to employment



Notes: Empirical hazards are estimated using a discrete-time logit model with B-splines. The dotted vertical line marks week 7, around which the initial increase in the hazard peaks. The dashed lines mark $t = 54$ and $t = 106$, chosen to align with the observed peaks around UI exhaustion rather than the nominal entitlements of 52 and 104 weeks.

that the unemployed graduates are very responsive to the exhaustion of UI and that the timing of benefit exhaustion is important for job-search behaviour.

The hazards provide less clear visual evidence of a response to differences in benefit levels across reform regimes. For the non-parent groups, which face a benefit reduction in the post-reform regime, the pre- and post-reform hazards are relatively similar up to exhaustion. For non-parents below age 30, the post-reform hazard is slightly higher throughout most of the UI period, indicating somewhat higher exits to employment before exhaustion. There is no clear kink or sudden change around the week-12 benefit reduction introduced in the post-reform regime that would indicate a strong response to the front-loaded benefit structure. It should also be noted that, since the post-reform benefit schedule changes benefit levels, the front-loaded structure, and potential benefit duration, the empirical hazards cannot separately identify the effect of each policy component.

After UI exhaustion, the hazard declines rapidly. This may partly reflect selection among those who remain unemployed after exhausting their UI. Individuals with relatively high job-finding probabilities are more likely to exit before or around exhaustion. The individuals who have not found employment after this point will likely have low job-finding probabilities. In addition, not all individuals are eligible for social assistance after exhausting their UI. Some individuals will exit their unemployment spell to states other than employment such as self-sufficiency or non-job-search benefits. Table 2 shows the distribution of exit states from unemployment.

Employment is the most common observed exit state for parents and non-parents aged 30 or above, while self-sufficiency accounts for a large share of exits among non-parents below age 30. This is important for interpreting the employment hazards, since exits to self-sufficiency and non-job-search benefits are treated as right-censored. The empirical hazards therefore describe exits to employment among individuals who remain at risk of an employment exit, not the full distribution of exits from unemployment. The large share of non-employment exits also supports the interpretation that the group remaining after UI exhaustion may be highly selected.

Table 2: Distribution of unemployment exit states

	Non-parents under 30		Non-parents 30+		Parents	
	<i>N</i>	%	<i>N</i>	%	<i>N</i>	%
Employment	45,826	41.2	8,837	46.8	14,682	50.6
Non-job-search benefits	13,265	11.9	3,019	16.0	5,979	20.6
Self-sufficiency	46,430	41.8	5,604	29.7	7,271	25.1
Censored	5,675	5.1	1,416	7.5	1,086	3.7

Notes: Non-job-search benefits refer to social benefits received during periods in which recipients are not expected to actively seek or take up employment. This category includes parental leave benefits, sickness benefits, educational support, and similar benefits.

Overall, the empirical hazards show strong duration dependence, a declining hazard after the initial transition phase, and a clear increase in job-search around UI exhaustion. These patterns motivate the dynamic search model with unobserved heterogeneity introduced in the next section.

3 The Dynamic Structural Job-Search Model

This section presents the dynamic structural job-search model used in the analysis. The model follows the UI job-search framework with unobserved heterogeneity in DellaVigna et al. (2017), while abstracting from the reference-dependent preferences in their main specification. Unemployed graduates are modelled as forward-looking and choose search intensity in response to the current and expected future benefit schedule. The model is used to explain the decline in employment-exit hazards from week 7 onward and the increase around UI exhaustion as seen in Figure 2. I first discuss the role of UI incentives and dynamic selection, then specify the model and Bellman equation, and finally present its solution and the construction of the aggregate model-implied hazard.

3.1 Theoretical Framework for Dynamic Job Search

In standard dynamic job-search theory à la Mortensen (1987), the job-search decision of an unemployed individual is modelled as a dynamic decision problem. In each period, the unemployed chooses search intensity, s_t , which increases the probability of finding employment but is costly. Optimal search intensity therefore balances the cost of search against the expected gain from employment. In this model, I assume that search intensity, s_t , is equal to the probability of exiting to employment in the following period. Model-implied search intensities can then be compared directly to the empirical hazards.

In a UI job-search model, unemployed job-seekers are forward-looking and respond to the full path of unemployment benefits. A future benefit reduction or UI exhaustion lowers the future value of remaining unemployed and therefore increases the incentive to search before the reduction occurs. A basic job-search model predicts an increasing exit hazard up to benefit exhaustion and a constant exit rate thereafter. As discussed by DellaVigna et al. (2017), this basic model cannot account for a declining hazard in the unemployment spell. I therefore introduce unobserved heterogeneity in search costs such that dynamic selection, through changing the composition of the remaining job-seekers, can allow the model to generate declining aggregate hazards.

3.2 Model Environment and the Value Function

Following DellaVigna et al. (2017), I assume that wages are fixed and that once an individual exits to employment, they remain employed at that wage indefinitely. I also assume hand-to-mouth consumption, meaning that all income received in a given period is consumed in that period. This assumption abstracts from saving and borrowing decisions, which may be a less restrictive abstraction for newly graduated UI recipients than for more established workers, as graduates are generally at the beginning of their working lives and may therefore have limited initial assets with which to smooth consumption during unemployment. Nevertheless, the assumption rules out potentially relevant consumption-smoothing behaviour in the model. I adopt it as a simplifying feature of the model, since, together with the fixed-wage assumption, it yields a closed-form solution for optimal search effort and substantially reduces the computational burden of estimation.

Income while unemployed is determined by the relevant benefit schedule given by which reform regime we are under. In the model, I approximate the benefit path by two levels: the UI benefit level b^{UI} received until UI exhaustion at duration T_1 , and expected post-UI income $b^{\text{post-UI}}$ received thereafter. In the actual post-reform system, non-parents receive a higher UI benefit during the first 12 weeks, followed by a lower benefit rate until exhaustion. In the chosen model setup, I abstract from this front-loaded structure and apply the lower post-week-12 benefit rate throughout the entire UI entitlement period. The model therefore identifies responses to the main UI benefit level and the timing of UI exhaustion, but not to the front-loaded benefit structure. This simplification is motivated by the absence of a clear hazard response around week 12 and by the results in Appendix B, where imposing the full front-loaded schedule does not improve the overall fit and worsens the fit around exhaustion. I therefore prefer the more stripped-down specification with one UI benefit level, which allows the model to focus more clearly on the effects of general benefit generosity and potential benefit duration.

After UI exhaustion, individuals who remain unemployed may receive social assistance if they satisfy the relevant eligibility requirements. Since not all individuals are eligible some will enter self-sufficiency where they do not receive any social transfer or have any employment income. I assume that if a person is self sufficient they receive zero income. Income after UI exhaustion can then be approximated by the expected post-UI transfer income:

$$b^{\text{post-UI}} = \rho b^{\text{SA}} + (1 - \rho) \cdot 0 = \rho b^{\text{SA}} \quad (1)$$

where b^{SA} is the social assistance benefit level and ρ is the observed share receiving social assistance after UI exhaustion. This approximation treats all individuals as facing the same expected post-UI income. In reality, individuals may know before UI exhaustion whether they are likely to be eligible for social assistance, which may affect their search incentives. Ideally, the model would therefore include separate eligibility types with different post-UI income. Due to time constraints, I abstract from this heterogeneity and use this common expected post-UI income level as an approximation. Under these assumptions, period-specific income is given by

$$y_t = \begin{cases} w, & \text{if employed,} \\ b^{\text{UI}}, & \text{if unemployed and } t < T_1, \\ b^{\text{post-UI}}, & \text{if unemployed and } t \geq T_1, \end{cases} \quad (2)$$

where w is the fixed wage and T_1 denotes the duration at which UI is exhausted.

Under the hand-to-mouth assumption, consumption equals income in each period, $c_t = y_t$. Period utility can therefore be written as $u(y_t)$. I assume that $u(\cdot)$ is increasing and concave, reflecting positive but diminishing marginal utility of consumption. The only choice variable is search intensity, $s_t \in [0, 1]$. In the model, I assume that s_t is equal to the probability of exiting to employment in the following period. The search-cost function, $\psi(s_t)$, is assumed to satisfy

$$\begin{aligned} \psi(0) &= 0, & \psi'(0) &= 0, \\ \psi'(s_t) &> 0, & \psi''(s_t) &> 0, & \text{for } s_t > 0. \end{aligned}$$

Thus, the search-cost function is increasing and strictly convex in search intensity. Given the income path, utility function, and search-cost function, the unemployed chooses search intensity in each period to maximize current utility and expected future value. This value is represented by the Bellman equation

$$V_t^U = \max_{s_t \in [0, 1]} \{u(y_t) - \psi(s_t) + \delta [s_t V^E + (1 - s_t) V_{t+1}^U]\}, \quad (3)$$

where $\delta \in [0, 1)$ is the discount factor.

3.3 Solving the Model

The model is solved by backward induction. Once UI is exhausted, the unemployed individual receives the constant post-UI income level $b^{\text{post-UI}}$ in all subsequent periods. The model therefore reaches a stationary post-exhaustion state. I first solve for the stationary values of employment and post-UI unemployment, and then move backwards through the finite UI period. The model solution follows the same general setup as in DellaVigna et al. (2017).

Since an individual who enters employment remains employed at the fixed wage w , the value of employment is stationary and given by

$$V^E = u(w) \sum_{j=0}^{\infty} \delta^j = \frac{u(w)}{1 - \delta} \quad (4)$$

After UI exhaustion, an unemployed individual receives $b^{\text{post-UI}}$ in every period. Thus, the choice of search effort is constant after this point in time. This steady state value of search intensity, s_{ss} , is given by the maximization problem

$$V_{ss}^U = \max_{s_{ss}} \frac{u(b^{\text{post-UI}}) - \psi(s_{ss}) + \delta s_{ss} V^E}{1 - \delta(1 - s_{ss})} \quad (5)$$

For an interior solution, the first-order condition for optimal search intensity at duration t is

$$\psi'(s_t^*) = \delta [V^E - V_{t+1}^U].$$

Since the search-cost function is strictly convex, its derivative is strictly increasing and therefore invertible (Kristensen & Møller, 2021, Theorem 3.10). Optimal search intensity can therefore be written as

$$s_t^* = (\psi')^{-1}(\delta [V^E - V_{t+1}^U]), \quad (6)$$

subject to $s_t^* \in (0, 1)$. See Appendix D for the full derivation. Given V^E and V_{ss}^U , the value of unemployment during the UI period is obtained by backward induction. I choose a terminal duration sufficiently far beyond UI exhaustion and set the unemployment value at this point equal to V_{ss}^U . From there, I work backward one period at a time, using the known continuation value V_{t+1}^U to determine optimal search intensity and the corresponding unemployment value. This procedure continues until the first period is reached.

With unobserved heterogeneity, the model produces a type-specific optimal search intensity, $s_{t,q}^*$, for each type q . The aggregate model-implied hazard is obtained by weighting these type-specific search intensities by the share of each type among the individuals still unemployed at duration t :

$$h_t^{\text{model}} = \sum_q \pi_{t,q} s_{t,q}^*, \quad (7)$$

where $\pi_{t,q}$ denotes the share of type q among the remaining unemployed population. The unemployed population is determined by updating and normalizing the type shares after each period.

4 Structural Estimation and Model Specification

This section describes how the structural model is estimated by matching model-implied hazards to the empirical hazards presented in Figure 2. I first present the Generalized Method of Moments (GMM) estimator. Then I describe the estimation setup and variation, specify the model functions and define the model inputs.

4.1 Generalized Method of Moments Estimator

I estimate the structural parameters using the GMM estimator as presented in DellaVigna et al. (2017), which chooses the parameter values that minimize the weighted distance between the

empirical hazards and the corresponding hazards predicted by the structural model. Let $\hat{m}$ denote the vector of empirical hazard moments and let $m(\theta)$ denote the corresponding vector of model-implied hazards for parameter vector θ . The GMM estimator is defined as

$$\hat{\theta} = \arg \min_{\theta} [\hat{m} - m(\theta)]' W [\hat{m} - m(\theta)], \quad (8)$$

where W is a weighting matrix. The empirical hazard moments are estimated using a logistic regression with B-splines. Because these hazards are estimated from a fitted logit model, I compute their covariance matrix $\hat{\Omega}$ using the delta method as described in Appendix C. For the GMM estimation, I use the diagonal weighting matrix

$$W = \text{diag}(\hat{\Omega})^{-1}. \quad (9)$$

This gives greater weight to hazards that are estimated more precisely, while simplifying the weighting procedure by not using the covariance between moments. Since the weighting matrix is diagonal rather than the full inverse covariance matrix, I use the general GMM sandwich formula (Newey & McFadden, 1994):

$$\widehat{\text{Var}}(\hat{\theta}) = (G'WG)^{-1}G'W\hat{\Omega}WG(G'WG)^{-1}, \quad (10)$$

where

$$G = \left. \frac{\partial m(\theta)}{\partial \theta'} \right|_{\theta=\hat{\theta}} \quad (11)$$

is the Jacobian of the model-implied moments with respect to the structural parameters. I approximate the Jacobian numerically using finite differences. For parameter j , the corresponding column $G_{\cdot j}$, of the Jacobian is computed as

$$G_{\cdot j} \approx \frac{m(\hat{\theta} + \varepsilon e_j) - m(\hat{\theta} - \varepsilon e_j)}{2\varepsilon}, \quad (12)$$

where e_j is a unit vector with one in position j and zero elsewhere, and ε is a small step size.

4.2 Model Specification and Estimation Strategy

The empirical moments are the estimated hazards of exit from unemployment to employment presented in Figure 2. Since the structural model is not set up to generate the sharp increase in hazards during the initial transition phase, I exclude moments for durations $t < 7$ from the estimation. Although the initial 6 weeks are not used in the structural estimation, a complete hazard path is required for the comparisons with reduced-form evidence and for the policy counterfactuals. I approximate the hazard path for weeks $t = [1; 6]$ by linear interpolation. This keeps the approximation simple and broadly reflects the increasing pattern observed in the empirical hazards during the initial transition phase. For each group, I set $h(1)$ equal to the average of the empirical pre- and post-reform hazards at $t = 1$. The hazards for $t = 2, \dots, 6$ are then obtained by linearly interpolating between this initial value and the model-implied hazard

at $t = 7$. Consequently, differences between simulated hazard paths during the initial six weeks should be interpreted as a result of this interpolation rather than as behaviour explained by the structural model.

I estimate the model separately for each of the three groups: non-parents below age 30, non-parents aged 30 or above, and parents. Within each group, I jointly use the pre- and post-reform hazards by stacking them into a single moment vector, while restricting the structural parameters to be constant across reform regimes. The 2023 reform changed the potential UI duration for all groups and reduced the UI benefit level for non-parents. For parents, the reform therefore provides variation only in potential benefit duration, while for non-parents it provides variation in both PBD and benefit levels. Since both dimensions changed simultaneously for non-parents, their separate effects cannot be cleanly distinguished from the reform variation alone. Using both regimes strengthens the identification of behavioural responses, but means that the reform cannot also be used for out-of-sample validation.

For each group, the structural model produces a corresponding vector of model-implied hazards $m(\theta)$, using the optimal search policy $s_{t,q}^*$, and the changing composition of unobserved search-cost types over the unemployment spell. No structural parameters are shared across the three groups, as I expect age and parental status to influence both job-search incentives and responses to expected changes in income. The moment vector contains 171 hazards for non-parents below age 30, 160 for non-parents aged 30 or above, and 162 for parents. The number differs across groups because weeks with too few observed exits are excluded. To reduce the risk of convergence to a local minimum, I estimate each model from several different initial parameter values and retain the solution with the lowest objective-function value.

I estimate the model using two utility specifications. The benchmark model uses standard log utility, while shifted-log utility is considered as an alternative specification. These two specifications were selected from a broader set of utility functions considered during model development, as they provided the most stable estimates and the closest fit to the empirical hazards. The benchmark log-utility specification is given by

$$u(c_t) = \ln(c_t). \quad (13)$$

This is a simple and relatively restrictive specification with no additional utility parameter. The alternative shifted-log specification is given by

$$u(c_t) = \ln(c_t - \bar{c}), \quad c_t > \bar{c}, \quad (14)$$

where $\bar{c}$ represents a subsistence consumption level. The economic intuition is that individuals require a basic level of consumption before deriving utility from the excess consumption $c_t - \bar{c}$. Since this gap must be positive, the $\bar{c}$ parameter has the constraint

$$\bar{c} < \min \{b^{\text{post-UI}}, b^{\text{UI}}, w\} = b^{\text{post-UI}}$$

Compared with log utility, the shifted-log specification is more flexible because it introduces the additional parameter $\bar{c}$.

I use the same functional form for the search-cost function as in DellaVigna et al. (2017):

$$\psi(s_t) = \kappa \frac{s_t^{1+\gamma}}{1+\gamma}, \quad (15)$$

where $\kappa > 0$ determines the level of search costs and $\gamma > 0$ determines the curvature of the search-cost function. Appendix D derives the closed-form solution for optimal search intensity s_t^* , under this specific search-cost function. Following DellaVigna et al. (2017), I introduce unobserved heterogeneity through the search-cost parameter κ . I include three discrete search-cost types in the model, with parameters

$$0 < \kappa_{\text{low}} < \kappa_{\text{med}} < \kappa_{\text{high}}.$$

The positivity restriction ensures that search effort is costly, while the ordering identifies the three unobserved types as low-, medium-, and high-search-cost individuals. The initial type shares π_q , are also estimated and restricted by

$$\pi_{\text{low}}, \pi_{\text{med}} \geq 0, \quad \pi_{\text{low}} + \pi_{\text{med}} \leq 1, \quad \pi_{\text{high}} = 1 - \pi_{\text{low}} - \pi_{\text{med}}.$$

These restrictions ensure that the shares are non-negative and sum to one. The type-specific hazards are aggregated using the evolving type shares among the remaining unemployed, as described in equation (7).

The model uses a group-specific fixed wage. I define these from data among individuals in my estimation sample who exit to employment. Benefit levels are approximated using group-specific averages of the institutional benefit rates before and after the reform, expressed in 2025 prices and weighted by the distribution of yearly UI entry cohorts. Benefit level $b^{\text{post-UI}}$ is the weighted average of social assistance and self sufficiency. The weight ρ , representing the share eligible for social assistance after UI exhaustion, is approximated from the estimation sample by comparing individuals who remain unemployed after exhaustion with those who exit to self-sufficiency around exhaustion. The model input parameters are reported in Table 3.

Table 3: Benefit-schedule parameters in the structural model

	Pre-reform			Post-reform		
	Non-parents under 30	Non-parents 30+	Parents	Non-parents under 30	Non-parents 30+	Parents
Mean wage (w)	8,020	7,867	7,732	8,020	7,867	7,732
UI rate (b^{UI})	3,522	3,522	4,039	2,358	2,978	3,932
Share of SA eligible (ρ)	0.67	0.71	0.71	0.58	0.65	0.68
Post-UI income ($b^{\text{post-UI}}$)	1,288	2,117	2,552	1,085	1,887	2,409
UI exhaustion (T_1)	106	106	106	54	54	54

Notes: Wages and benefit rates are weekly amounts in DKK. The parameter ρ is the estimated share eligible for social assistance after UI exhaustion, and $b^{\text{post-UI}} = b^{\text{SA}}\rho$. The exhaustion points $T_1 = 106$ and $T_1 = 54$ are chosen to align with the observed exhaustion spikes rather than the nominal pre- and post-reform entitlements of 104 and 52 weeks.

5 Model Fit, Structural Results, and Policy Counterfactuals

This section presents the results from estimating the structural model. I first evaluate the fit and parameter estimates of the benchmark log-utility specification. I then compare these results with the alternative shifted-log specification, which allows additional flexibility in the utility function. Using the benchmark model, I compare selected model-implied policy effects with existing reduced-form evidence to strengthen the validity of my model. Lastly, I produce policy counterfactuals to examine the model-implied policy implications for different changes to the UI system.

5.1 Model Fit and Structural Parameter Estimates

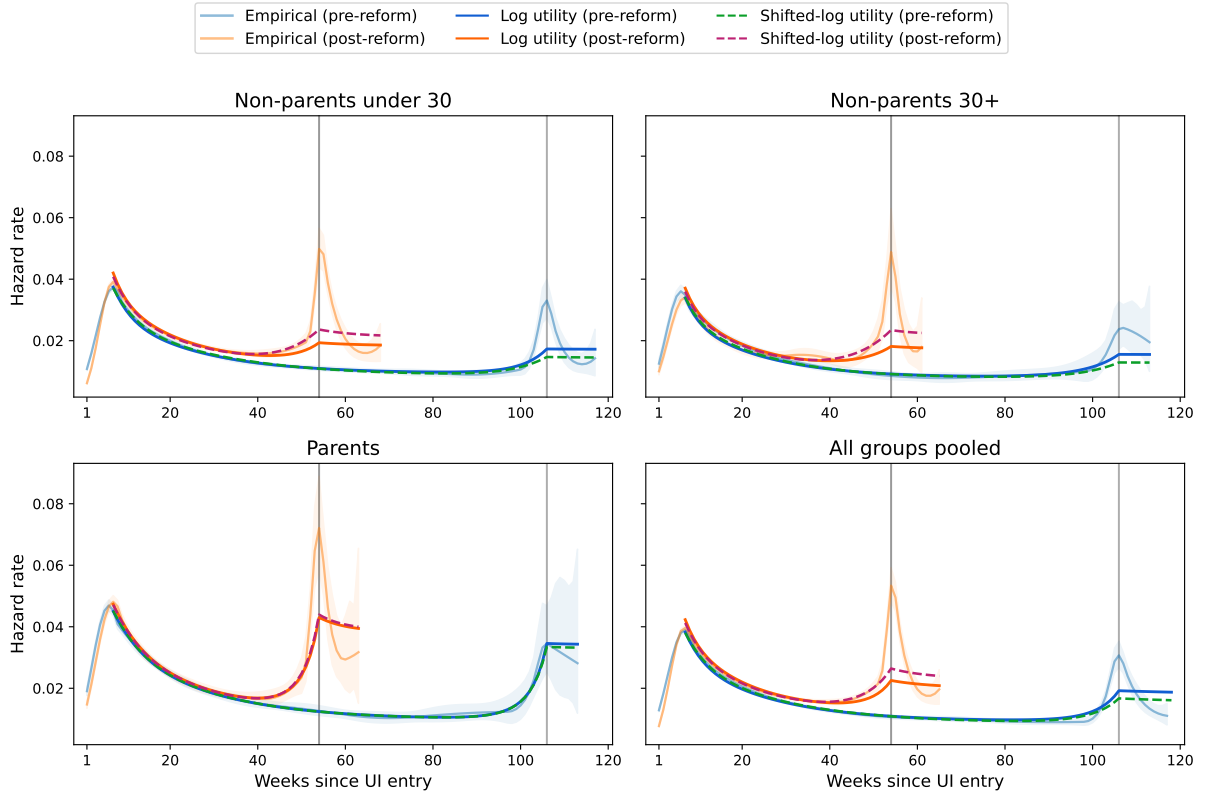
The fit of the benchmark log-utility specification is shown in Figure 3. Overall, the model fits the declining hazards from week 7 to around week 45 and the increase as UI exhaustion approaches. However, it has difficulty reproducing the magnitude of the sharp exhaustion spike, especially for the two non-parent groups. This suggests that the model is able to capture the gradual decline in job-search intensities over the unemployment spell, but is less successful in capturing the sharp behavioural response at benefit exhaustion.

The estimated parameters under log utility are reported in Table 4. The log-utility parameters are generally estimated with relatively small standard errors, particularly the discount factor, curvature parameter, and type shares. The search-cost parameters are less precisely estimated, with uncertainty increasing for the higher-cost types and especially for κ_{high} among non-parents. Even so, the estimated κ -values remain sufficiently separated within each group to support a clear distinction between low-, medium-, and high-cost types. All three types also remain active job seekers throughout the unemployment spell, so the model does not generate a group that never searches for employment. This can be seen in the type-specific optimal search intensities shown in Appendix E.

The model estimates a large share of high-cost types, ranging from 41–53%, making this the largest type for all groups. For the non-parent groups, the estimated search costs differ substantially across types. The low- and medium-cost types exit more quickly, causing the remaining unemployed population to become increasingly dominated by the already large high-cost group. This may be a thing that limits the aggregate response around UI exhaustion, as the composition of the unemployed group at this point will be largely individuals with lower job-finding probabilities. For parents, the estimated search costs are less widely separated, so the composition of the unemployed population changes less sharply over the spell. A larger share of the lower-cost types therefore remains near UI exhaustion and can contribute to the stronger aggregate response. This difference in dynamic selection may partly explain why the model reproduces the exhaustion spike more closely for parents than for non-parents.

In general, the parent group seems to be the best captured by the structural model. It provides the best fit to the empirical hazards, and the results are very similar across the log-utility and shifted-log specifications. Beside the dynamic selection argument, another possible explanation is that parents are affected mainly by the change in potential benefit duration, while their income loss at exhaustion remains similar across reform regimes. For non-parents,

Figure 3: Fit of the estimated models under both utility specifications



Notes: Vertical lines mark UI exhaustion. The model is estimated using moments for $t \geq 7$, and model-implied hazards are shown from this point onward. The pooled model-implied hazard is constructed by aggregating the hazards of the three groups.

lower post-reform UI benefits reduce the gap between UI and expected post-UI income. The model would therefore imply both a response to lower benefits throughout the spell and a weaker income loss at exhaustion. In the empirical hazards, however, the additional response to the benefit-level reduction is not very pronounced, while the exhaustion response remains strong. This may make the non-parent hazards more difficult to fit and could help explain why the parent group is captured more closely.

One feature that stands out across both utility specifications is that the estimated discount factors are relatively low for a model with weekly periods. This implies substantial discounting of future utility. One possible interpretation is that the discount factor partly captures features of the hazard profile that the model does not fully explain, particularly the sharp response around UI exhaustion. The estimates should therefore not be interpreted as precise measures of individual time preferences.

Relative to the benchmark log specification, shifted-log utility improves the fit around post-reform UI exhaustion for the two non-parent groups. However, it produces a lower exhaustion spike in the pre-reform regime. Although the pre-reform fit is worse, this alternative specification may better capture the shift in the spike level between the pre- and post-reform periods observed in the empirical hazards. The improved non-parent fit under shifted-log utility comes at the cost of less convincing structural parameter estimates. For both non-parent groups, the

estimated subsistence parameter, $\bar{c}$, is very close to its upper bound, given by expected post-UI income, $b^{\text{post-UI}}$. Rather than identifying a plausible subsistence consumption level, the parameter appears to be used mainly to fit the sharp exhaustion response. The shifted-log parameters are also less precisely estimated than the log-utility parameters. This is especially clear for the curvature parameter (γ) and the search-cost parameters (κ), whose standard errors are large relative to the point estimates.

Table 4: Estimated structural parameters

Benchmark: Log utility			
Parameter	Non-parents under 30	Non-parents 30+	Parents
δ	0.83 (0.05)	0.89 (0.05)	0.83 (0.03)
γ	1.29 (0.09)	0.71 (0.10)	0.40 (0.04)
κ_{low}	15.47 (5.38)	8.53 (4.43)	5.28 (1.34)
κ_{medium}	119.21 (44.40)	36.19 (16.15)	9.30 (1.18)
κ_{high}	1,672.48 (858.46)	199.67 (130.34)	20.56 (3.70)
π_{low}	0.20 (0.02)	0.18 (0.05)	0.24 (0.10)
π_{medium}	0.30 (0.01)	0.29 (0.02)	0.35 (0.07)
π_{high}	0.50 (0.02)	0.53 (0.04)	0.41 (0.04)
Alternative: Shifted-log utility			
Parameter	Non-parents under 30	Non-parents 30+	Parents
δ	0.85 (0.02)	0.88 (0.04)	0.83 (0.10)
γ	2.32 (5.02)	1.83 (10.10)	0.61 (1.69)
$\bar{c}$	-1,084.20 (15.23)	-1,886.29 (22.77)	-1,912.99 (118.85)
κ_{low}	147.30 (85.21)	41.47 (45.37)	10.46 (9.78)
κ_{medium}	3,920.94 (2,998.40)	943.38 (1,403.50)	23.94 (24.42)
κ_{high}	358,574.59 (475,806.48)	62,992.16 (183,989.12)	79.57 (146.10)
π_{low}	0.20 (0.03)	0.17 (0.04)	0.24 (0.06)
π_{medium}	0.32 (0.01)	0.31 (0.02)	0.36 (0.04)
π_{high}	0.48 (0.02)	0.52 (0.03)	0.40 (0.03)

Notes: Standard errors are reported in parentheses.

The improved fit under shifted-log utility should therefore be interpreted cautiously. The additional flexibility helps the model capture the sharp exhaustion response in the post-reform hazards, but the corresponding structural parameters are not estimated precisely enough to support a strong economic interpretation. I therefore retain log utility as the benchmark specification, while treating shifted-log utility as an alternative specification that illustrates the trade-off between model fit and structural interpretability.

5.2 Comparison with Existing Reform Evidence

Since the pre- and post-reform hazards are both used in the structural estimation, the 2023 reform cannot serve as an out-of-sample validation exercise. Instead, I compare policy effects implied by the benchmark log-utility model with reduced-form estimates from existing evaluations of the 2017 and 2023 reforms. These comparisons provide a plausibility check for whether the structural model produces effects of a similar direction and magnitude to previous empirical evidence.

The 2023 Reform: Comparison with Sørensen

Sørensen (2025) evaluates the 2023 reform to the UI graduate scheme by considering its two main policy changes separately. First, the effect of the benefit-level reduction for non-parents is estimated using a difference-in-differences design, where non-parents constitute the treatment group and parents, whose benefit level was unchanged, serve as the control group. Second, the effect of the reduction in potential benefit duration is examined for parents, who experienced a shortening of PBD without a simultaneous change in their UI benefit level. Following the logic of policy-relevant treatment effects in Vytlacil (2005), I construct model-implied analogues of treatment effects by comparing the hazards predicted by the structural model under alternative benefit regimes.

To isolate the model-implied effect of the benefit-level reduction for non-parents, I compare the predicted post-reform hazard with a counterfactual hazard that retains the post-reform PBD but replaces the post-reform benefit level with the pre-reform level. For each non-parent group, the model-implied effect is

$$\xi_g^{\text{level}}(t) = h_g^{\text{post}}(t) - h_g^{\text{cf, level}}(t), \quad (16)$$

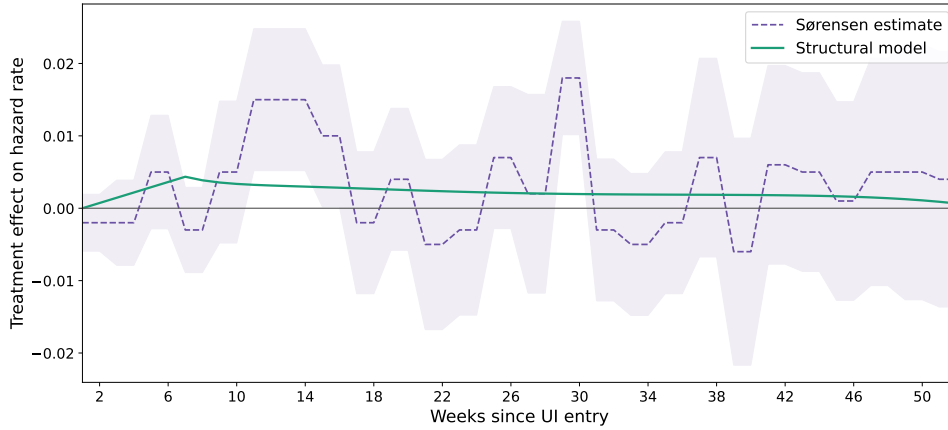
where $h_g^{\text{post}}(t)$ is the hazard under the post-reform benefit schedule and $h_g^{\text{cf, level}}(t)$ is the counterfactual hazard under pre-reform benefit levels and post-reform PBD. The two non-parent groups are aggregated using their sample shares to obtain a pooled non-parent effect comparable to the estimate reported by Sørensen (2025).

Figure 4 shows that the model-implied effect of lowering benefit levels is generally positive and of a similar magnitude to Sørensen’s estimates. The model does not reproduce the response around the 12-week benefit reduction, since the model specification abstracts from the front-loaded component of the post-reform schedule. Moreover, Sørensen’s estimates are imprecise for most durations. The comparison therefore provides only suggestive evidence, but the model-implied response is broadly consistent with the direction of the reduced-form estimates.

To isolate the effect of shorter PBD, I compare the predicted post-reform hazard for parents with a counterfactual hazard that retains the post-reform benefit level but restores the pre-reform PBD:

$$\xi_{\text{parents}}^{\text{duration}}(t) = h_{\text{parents}}^{\text{post}}(t) - h_{\text{parents}}^{\text{cf, duration}}(t). \quad (17)$$

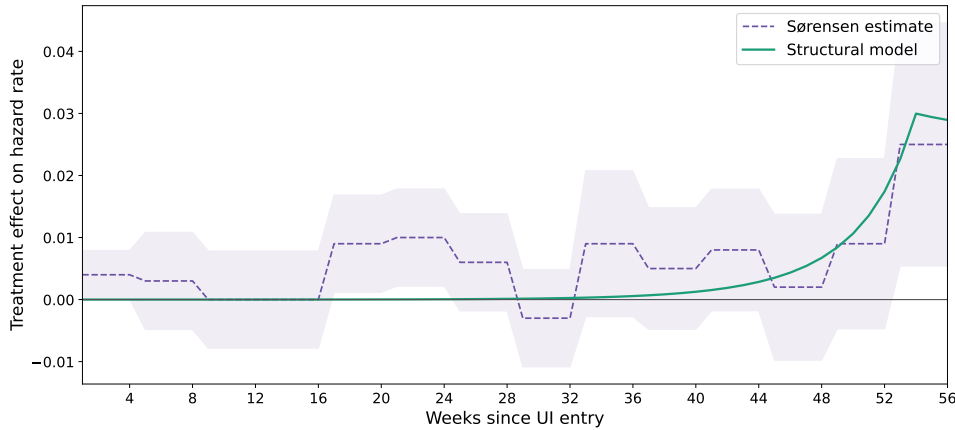
Figure 4: Treatment effect of the benefit-level reduction: comparison with Sørensen



Notes: Treatment effect hazards and standard errors are taking from Sørensen (2025): Table 3, column (1).

Figure 5 shows that both the structural model and Sørensen’s reduced-form estimates imply a positive effect of the shortening of PBD on employment-exit hazards around the new exhaustion point. The model-implied effect is zero at earlier durations and rises sharply near exhaustion, whereas Sørensen’s estimates suggest a smaller positive effect spread more evenly across the unemployment spell, although these estimates are generally imprecise and the effect is not statistically significant at most durations. Around exhaustion, the structural model predicts an increase in the hazard that follows the reduced-form estimate quite closely. Thus the structural model captures the direction and timing of the duration effect, as well as a similar response around exhaustion as found by Sørensen (2025).

Figure 5: Treatment effect of shorter potential benefit duration: comparison with Sørensen



Notes: Treatment effect hazards and standard errors are taking from Sørensen (2025): Table 4, column (5).

Overall, the comparison with Sørensen (2025) suggests that the benchmark model produces policy responses in the same general direction as the reduced-form evidence, while also revealing a limitation: the model cannot reproduce responses to the initial front-loaded benefit reduction.

The 2017 Reform: Comparison with STAR

STAR (2022) evaluates the 2017 reform, which reduced the graduate UI benefit rate for non-parents from 82.0 percent to 71.5 percent of the maximum UI benefit rate, while leaving the benefit rate for parents unchanged. As in Sørensen’s analysis of the benefit-level component, the effect is estimated using a difference-in-differences design, where non-parents constitute the treatment group and parents serve as the control group. The analysis reports the implied elasticity of unemployment duration with respect to the benefit level change and the corresponding change in annual full-time-equivalent unemployed.

To construct the corresponding model-implied effect, I use the pre-2023 regime, under which potential benefit duration remains two years, and compare the model prediction with a counterfactual in which non-parents receive the higher pre-2017 benefit level. All other model inputs are held fixed. The counterfactual benefit level is obtained by scaling the pre-2023 non-parent UI benefit by $(0.82/0.715)$. For a given model-implied hazard path, the survival probability at duration (t) is

$$S(t) = \prod_{j=0}^{t-1} (1 - h(j)), \quad (18)$$

and expected unemployment duration is

$$\mathbb{E}[T] = \sum_{t=0}^T S(t). \quad (19)$$

The model-implied elasticity is computed as

$$\varepsilon = \frac{\Delta \mathbb{E}[T] / \mathbb{E}[T]_{\text{pre-2017}}}{\Delta b^{\text{UI}} / b^{\text{UI}}_{\text{pre-2017}}}. \quad (20)$$

where $\Delta \mathbb{E}[T] = \mathbb{E}[T]_{\text{post-2017}} - \mathbb{E}[T]_{\text{pre-2017}}$ and $\Delta b^{\text{UI}} = b^{\text{UI}}_{\text{post-2017}} - b^{\text{UI}}_{\text{pre-2017}}$.

Table 5 shows that the benchmark model implies an elasticity of 0.41, compared with 0.44 in STAR (2022). The model-implied change in annual full-time-equivalent unemployment is quite close to the STAR estimate and lies within its reported confidence interval. This comparison is reassuring for the model’s prediction of the aggregate effect of changes in UI benefit levels. However, it does not test whether the model reproduces the full duration profile of the response, and the reduced-form estimate itself is subject to considerable uncertainty.

Table 5: Employment effect of the 2017 benefit reduction: comparison with STAR (2022)

	Elasticity (ε)	Δ annual FTE	95% CI
STAR (2022)	0.44	−770.5	[−1400.6; −128.5]
Estimated model	0.41	−803.7	—

Note: FTE effect is computed as $\Delta \text{FTE} = N \cdot \Delta E / 52.14$, where $N = 13,589$ is the annual non-parent graduate cohort used for FTE estimates in STAR (2022).

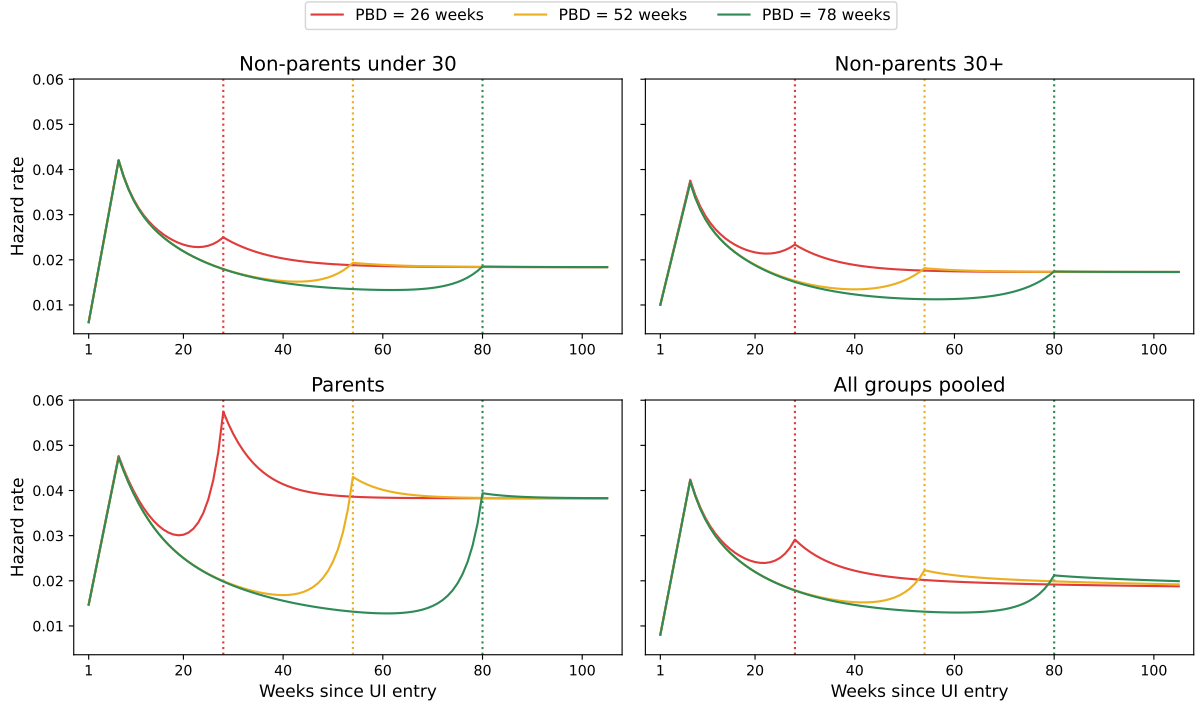
Taken together, the comparisons with Sørensen (2025) and STAR (2022) provide some support for the external plausibility of the benchmark model. The model-implied policy effects are generally consistent with the direction and magnitude of the reduced-form evidence, and the aggregate benefit-level effect is particularly close to the estimate reported by STAR.

5.3 Policy Counterfactuals for Graduate UI Design

I use the benchmark log-utility specification to examine job-search behaviour under alternative graduate UI schedules. The two policy levers are potential benefit duration and UI benefit levels. I find that changes in PBD mainly shift the timing of the response around exhaustion, while benefit-level changes affect search intensities throughout the UI period.

Figure 6 shows model-implied hazards under three lengths of PBD, with benefit levels fixed at the post-reform levels. For all groups, the increase in the hazard moves to the new exhaustion point, while the hazard paths remain similar until exhaustion approaches. Changes in PBD therefore mainly affect search intensity close to exhaustion. This is consistent with the relatively low estimated discount factors, which imply that the unemployed respond most strongly when the future benefit reduction is imminent. Parents show the strongest model-implied exhaustion response. This, however, should be interpreted cautiously, as it may reflect genuine differences in responsiveness across groups, but it may also reflect that the parent hazard profile is better captured by the model.

Figure 6: Counterfactual hazards under alternative potential benefit durations

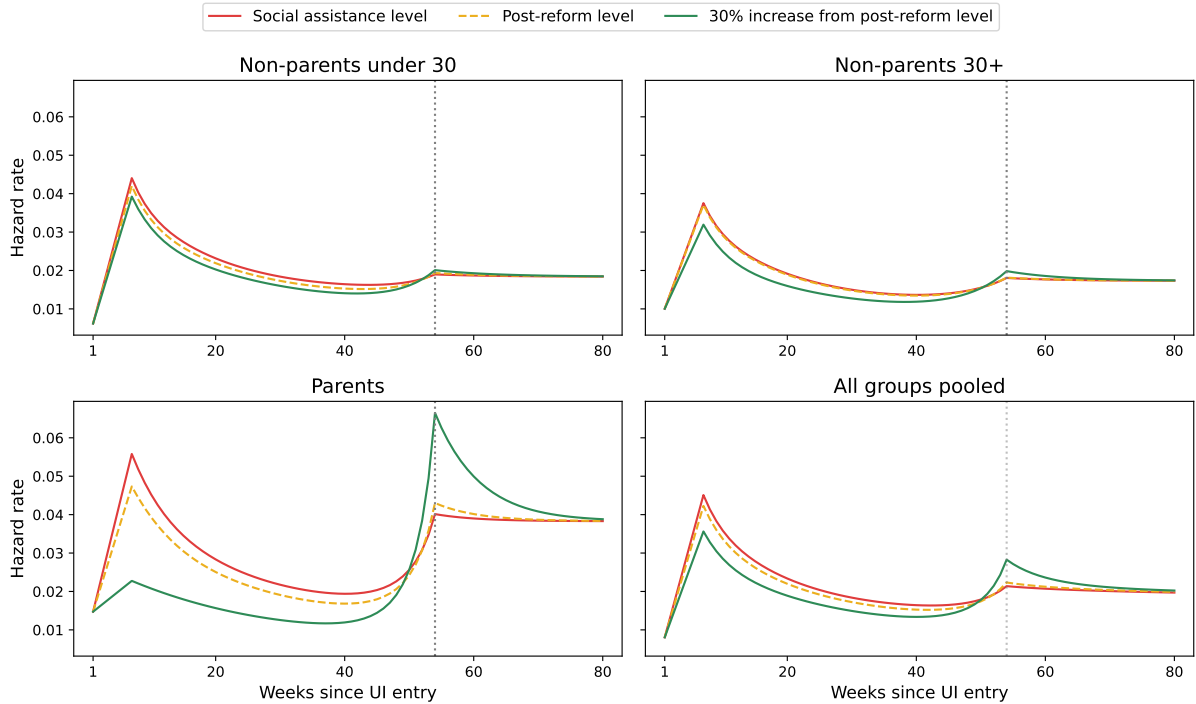


Notes: UI benefit levels are held fixed at their post-reform values. The vertical lines mark the UI exhaustion week under each counterfactual potential benefit duration.

Figure 7 shows model-implied hazards under different UI benefit levels, with PBD fixed at the post-reform level. Benefit changes are defined relative to each group's post-reform UI benefit level. The counterfactual benefit level is bounded by each group's social assistance level, since reducing UI below this level would not be meaningful within the current institutional setting. This is particularly restrictive for non-parents aged 30 or above, whose UI benefit is only slightly above this lower bound. As a result, I can only examine a small range of reductions in UI benefits.

Lower benefit levels increase job-search intensity throughout the UI entitlement period, while higher benefits reduce it. This reflects the standard search-incentive mechanism: when unemployment benefits are lower, the value of remaining unemployed falls and search effort increases. Since the counterfactual changes are defined relative to each group's own post-reform benefit level, these differences should not be interpreted as directly comparable measures of responsiveness across groups. Instead, it shows how the estimated model predicts each group to respond to changes around its current institutional setting. Within this setting, the response is modest for non-parents below age 30, larger for non-parents aged 30 or above, and strongest for parents. Figure 7 also shows that higher benefit levels can generate a larger spike around UI exhaustion. This reflects dynamic selection, as higher benefits reduce early exits and change the composition of the unemployed population remaining at exhaustion, leading to a larger aggregate response.

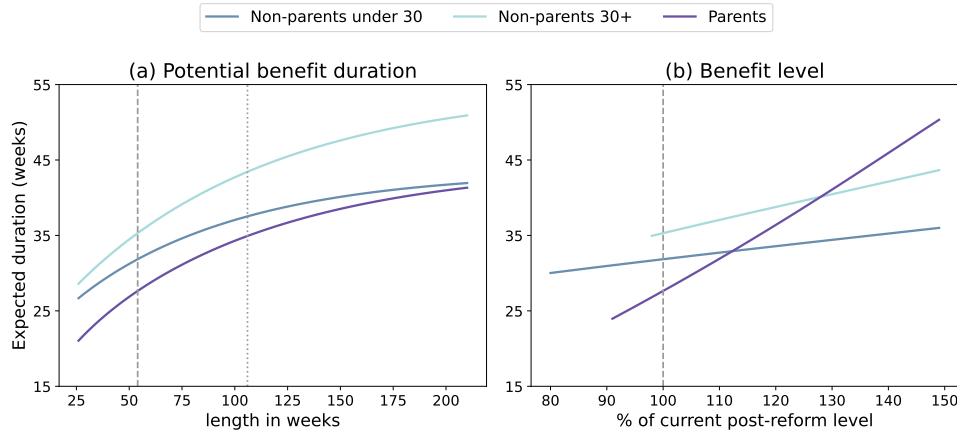
Figure 7: Counterfactual hazards under alternative benefit levels



Notes: Benefit levels are increased relative to each group's post-reform level and reduced to the group-specific social-assistance level, as this serves as the lower bound. Potential benefit duration is fixed at the post-reform value, marked by the vertical line.

Figure 8 summarizes the counterfactual hazards in terms of expected unemployment duration. Panel (a) shows that expected duration increases with potential benefit duration for all three groups, but at a decreasing rate. Parents and non-parents aged 30 or above respond similarly, as indicated by the comparable shape and slope of their curves, while non-parents below age 30 are less responsive, as shown by the flatter curve. Panel (b) shows that expected duration also increases with the UI benefit level, with the largest response among parents and only limited effects for non-parents below age 30. Although the latter group's benefit level can be reduced further before reaching the lower bound of the social assistance benefit level, the model predicts that such reductions would have relatively small effects on unemployment duration.

Figure 8: Expected unemployment duration under alternative UI schedules



Notes: Panel (a) varies potential benefit duration; the dashed and dotted lines mark the post- and pre-reform durations, respectively. Panel (b) varies UI benefit levels relative to the post-reform baseline, marked by the vertical line. Benefit reductions are not shown below each group's social-assistance level.

Taken at face value, the model implies that shorter potential benefit duration and lower UI benefit levels reduce expected unemployment duration and increase labour supply, with the largest labour-supply response to such reductions occurring among parents. These results should not, however, be interpreted as a complete policy evaluation. The model captures only direct effects on exits to employment and expected unemployment duration, while abstracting from other relevant outcomes, including income insurance, consumption losses, child welfare, and job-match quality. These omissions are especially important for parents, since lower benefits may reduce household income in ways that adversely affect children and potentially have long-term consequences. More generally, faster entry into employment is not necessarily the only policy objective, since more generous UI may allow longer search and potentially improve job matches. The counterfactuals should therefore be interpreted as model-implied employment responses to alternative UI schedules rather than recommendations for the optimal design of graduate UI.

6 Model Limitations and Directions for Further Research

The preceding results show that the model captures important features of unemployed graduate’s job-search behaviour and produces plausible responses to changes in the UI schedule. At the same time, the remaining gaps between the model and the empirical hazards point to several limitations and possible extensions.

The most glaring limitation is its inability to fully reproduce the magnitude of the exhaustion spike, especially for the two non-parent groups. This is particularly relevant because non-parents below age 30 constitute the largest group in the graduate UI system. One possible extension to combat this would be to add reference dependence, as in DellaVigna et al. (2017), which could potentially help generate sharper responses around UI exhaustion.

Another possible explanation for the lacking fit of the exhaustion spike is that the groups are more heterogeneous than the current model allows for. Unemployed graduates differ substantially in education types, expected wages, labour-market opportunities, and the types of jobs they are willing to accept. One way to accommodate this would be to extend the model with reservation wages, so that individuals differ not only in search costs but also in the wage offers they are willing to accept. Another extension would be to include more observed heterogeneity by education type. I briefly explored this possibility, but splitting the sample by education type leaves too little variation in later weeks and leads to small estimation samples, especially once the data are also divided by reform regime and benefit group. An alternative approach that would make it possible to include education types, would be to estimate the structural model directly on the individual-level data using maximum likelihood like in Lentz (2009). This would make it possible to include richer observed heterogeneity like education type without having to construct a separate empirical hazard curve for each subgroup.

A second limitation is that the model does not account for the sharp increase in employment exits during the first weeks of unemployment. This part of the hazard profile is excluded from the structural estimation, since the model is not designed to capture the transition phase immediately after UI entry. In the model, search intensity maps directly into the probability of entering employment and the empirical hazards reports the week in which employment actually begins. One possible interpretation of the early increase is that there is a delay between search activity and realized job starts. Even if unemployed graduates begin searching immediately after graduation, transitions into employment only materialize after applications have been sent, interviews have taken place, and actual start dates have been reached. A model without this delay mechanism may therefore predict too direct a link between current search effort and subsequent employment exits. A natural direction for further work would be to introduce a delay between search effort and realized employment. Instead of letting search intensity translate directly into employment, search effort could be modelled as generating successful applications or job contacts. These would not necessarily lead to employment immediately, but would accumulate over time and increase the probability of entering employment in later periods. This mechanism could help generate a low initial employment-exit hazard followed by an increase over the first weeks of unemployment. If newly unemployed graduates start with no active applications or job contacts, the probability of realized employment would initially be low even if they search

actively. As search effort accumulates into applications, the probability of entering employment would rise. The exact implementation of this mechanism is beyond the scope of this thesis but provides a natural direction for further research seeking to model the full hazard profile of unemployed graduates.

7 Conclusion

This thesis applies a dynamic structural job-search model to examine job-search behaviour among newly graduated UI recipients in Denmark. From week 7 onward, the benchmark model captures the decline in employment-exit hazards and the shift in the timing of the exhaustion response following the 2023 reform. The model is less successful in reproducing the sharp magnitude of the exhaustion spike, particularly for non-parents. The shifted-log specification improves this part of the fit, but only at the cost of less precisely estimated and less convincing structural parameters. This trade-off supports the use of log utility as the benchmark. The comparisons with existing reduced-form evidence provide some support for the plausibility of the model-implied responses, although uncertainty in the reduced-form estimates and differences in empirical design limit how strongly these comparisons can be interpreted.

The counterfactuals show that the two main policy levers operate differently. Changes in potential benefit duration primarily move the timing of the response around exhaustion, whereas changes in benefit levels affect job-search intensity throughout the UI period. Both shorter benefit duration and lower benefit levels reduce expected unemployment duration within the model, but these results capture employment responses only. They do not provide a complete assessment of the welfare consequences of alternative UI policies.

Overall, the thesis shows that a relatively simple structural search model can account for the medium-run and to some degree the exhaustion-related part of graduate job-search behaviour. At the same time, it shows where the presented model is incomplete. The model provides a useful framework for analysing how graduate UI benefit schedules affect job search after the initial transition phase, while future research could extend the framework to account for delays from job-search frictions and richer heterogeneity among newly graduated job seekers.

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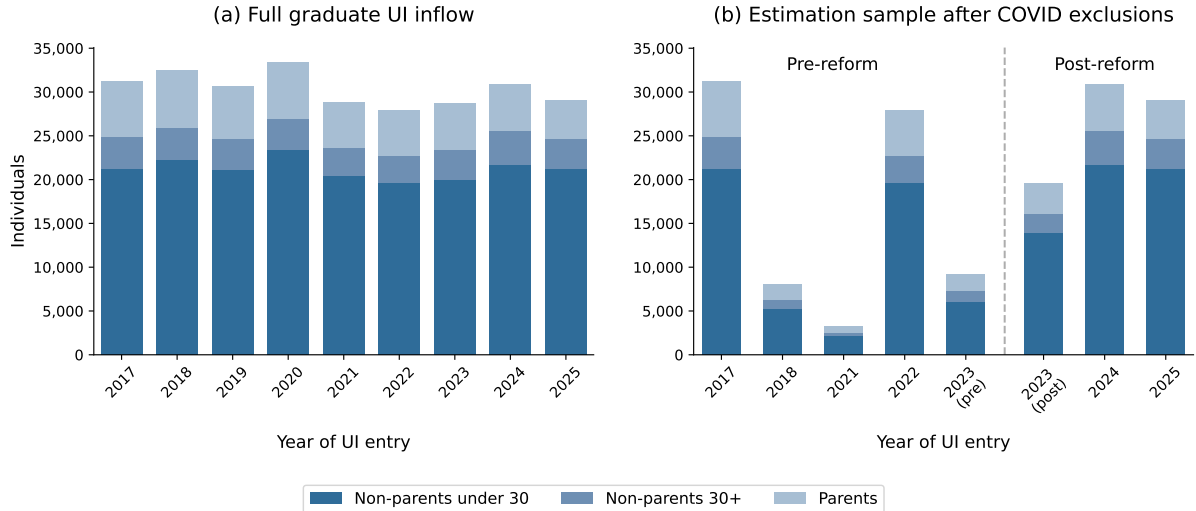
Appendix A: Exclusion of COVID-Affected Cohorts

The full sample contains 273,200 graduate UI entries between January 2017 and December 2025. Panel (a) of Figure 9 shows that the yearly inflow into graduate UI is relatively stable over this period. For the structural estimation, I exclude periods affected by the COVID-19 pandemic and the associated labour-market disruptions. The exclusions are based on a visual inspection of empirical employment-exit hazards estimated separately by UI-entry cohort. I exclude UI spells starting in 2019 and 2020, as well as nearby cohorts whose hazard profiles suggest that their unemployment spells were still affected by the pandemic. The 2019 cohort is excluded even though these spells began before the lockdowns, because the pre-reform potential benefit duration was two years. This means that the later part of many of these unemployment spells overlaps with the COVID-19 period.

The resulting pre-reform estimation period consists of 1 January 2017–28 February 2018 and 1 September 2021–30 April 2023, while the post-reform period runs from 1 May 2023 to 31 December 2025. After applying the exclusions, the estimation sample contains 159,090 graduate UI entries, of which 79,512 are observed under the pre-reform regime and 79,578 under the post-reform regime.

Panel (b) of Figure 9 shows the yearly inflow included in the estimation sample. The lower inflow in the excluded years follows mechanically from the sample restrictions, while the relative composition of the three benefit groups (non-parents under 30, non-parents 30+ and parents) remains broadly stable. The exclusions therefore reduce the number of observed spells but do not substantially change the group composition of the estimation sample.

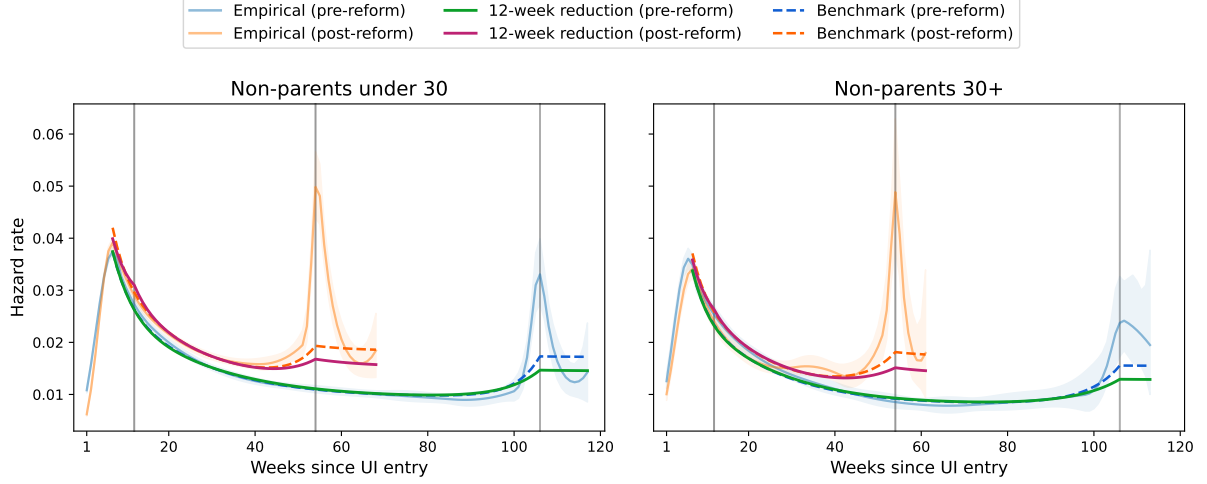
Figure 9: Yearly inflow into graduate unemployment insurance



Note: Panel (a) shows the yearly inflow into graduate UI of the full sample, while panel (b) shows the estimation sample. The pre-reform period covers entries in the period 1 January 2017–28 February 2018 and 1 September 2021–30 April 2023; the post-reform period covers entries in the period 1 May 2023–31 December 2025

Appendix B: Results from the Structural Model with a Week-12 Benefit Reduction

Figure 10: Model fit with the week-12 benefit reduction



Notes: The week-12 reduction specification incorporates the front-loaded post-reform benefit schedule, with UI benefits reduced after 12 weeks. The benchmark model, which abstracts from this reduction, is shown for reference. The first vertical line marks week 12, while the remaining lines mark pre- and post-reform UI exhaustion.

Table 6: Estimated structural parameters

Parameter	Non-parents under 30	Non-parents 30+
δ	0.87 (0.04)	0.92 (0.05)
γ	1.73 (0.14)	0.96 (0.18)
κ_{low}	39.44 (16.08)	13.06 (8.37)
κ_{medium}	604.82 (300.27)	90.46 (65.81)
κ_{high}	18,368.51 (13,320.51)	877.66 (1,007.64)
π_{low}	0.21 (0.02)	0.19 (0.04)
π_{medium}	0.29 (0.02)	0.29 (0.02)
π_{high}	0.49 (0.02)	0.52 (0.04)

Notes: Estimates are from the log-utility specification that incorporates the front-loaded post-reform UI schedule with a benefit reduction after 12 weeks. Standard errors are reported in parentheses.

Appendix C: Estimation of Empirical Hazards Using a Discrete-Time Logit Model with B-Splines

I estimate the empirical hazards of exit from unemployment to employment using the register-based unemployment spells in the estimation sample. The event of interest is exit to employment, while all other exits are treated as censored. To obtain smooth empirical hazard curves, I estimate a discrete-time logit model in which unemployment duration enters flexibly through B-spline basis functions. This approach follows the semi-parametric discrete-time event-history framework in Berger and Schmid (2017), where discrete-time hazards are modelled using binary response models and duration dependence is allowed to enter flexibly. The use of spline terms is motivated by Carter and Signorino (2010), who show how splines can be included in logistic regression models to flexibly capture how event probabilities change with duration. In my setting, these event probabilities are the weekly probabilities of exiting from unemployment to employment. Using this method, I avoid using a fully non-parametric approach, where I would include a separate dummy for each unemployment week, which creates noisy hazards.

Let E_{it} denote an indicator for whether individual i exits unemployment to employment in week t , and let D_i be the individual's exit week. The empirical hazard is given by the probability of exiting to employment conditional on being unemployed:

$$h_t = \Pr(E_{it} = 1 \mid D_i \geq t) = \Lambda(B(t)\alpha) = \Lambda\left(\sum_{j=1}^J \alpha_j B_{j,d}(t)\right), \quad (21)$$

where $\Lambda(\cdot)$ is the logistic function and

$$B(t) = (B_{1,d}(t), \dots, B_{J,d}(t)).$$

Here, $B_{j,d}(t)$ denotes the j 'th B-spline basis function of degree d , evaluated at duration t . The strength of using basis functions with B-splines is that it provides local flexibility while still having a curve that is connected at the knots. The basis function is defined recursively using the Cox-de Boor recursion formula (de Boor, 2001, Ch. IX):

$$B_{j,d}(x) = \frac{x - t_j}{t_{j+d} - t_j} B_{j,d-1}(x) + \frac{t_{j+d+1} - x}{t_{j+d+1} - t_{j+1}} B_{j+1,d-1}(x). \quad (22)$$

The degree-zero basis function is defined as

$$B_{j,0}(x) = \begin{cases} 1, & t_j \leq x < t_{j+1}, \\ 0, & \text{otherwise.} \end{cases} \quad (23)$$

Higher-degree B-splines are constructed recursively as weighted combinations of lower-degree B-splines. The knots t_j determine the intervals over which the basis functions are active. The covariance matrix for the fitted hazards is approximated using the delta method (Oehlert, 1992):

$$\widehat{\Omega} = D \widehat{\text{Var}}(\widehat{\alpha}) D', \quad (24)$$

where $\hat{\Omega}$ is the covariance matrix of the fitted hazards and D is the Jacobian. Let B denote the matrix containing the values of the B-spline basis functions for each unemployment week. Then

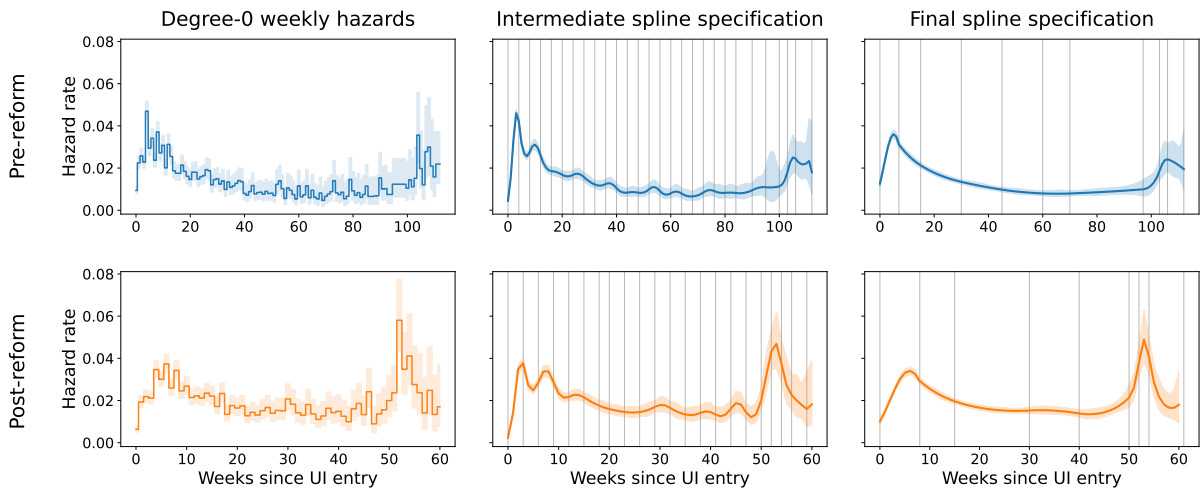
$$D = \left. \frac{\partial g(\alpha)}{\partial \alpha'} \right|_{\alpha=\hat{\alpha}} = \text{diag}(\hat{h}_t(1 - \hat{h}_t)) B.$$

This follows since, for each week t ,

$$\frac{\partial \hat{h}_t}{\partial \alpha'} = \hat{h}_t(1 - \hat{h}_t)B(t).$$

I use quadratic splines, as this provides smooth hazard curves that are flexible enough to capture the main features in the data while being comparable to the model-implied hazards from the structural model. The number of knots involve a trade-off between variance and bias. The knots are selected using the so-called "ocular" diagnostic described in Carter and Signorino (2010), where knot locations are determined by visual examination. The process of calibrating the knot placement is done by first using degree-0 B-splines with a knot in every week. In Figure 11 this is shown for non-parents over the age of 30. This creates noisy hazards but is a good reference point for the general movement in the hazards. Then the logistic regression is made using degree-2 B-splines and tightly placed knots. In Figure 11, we see that this intermediate step gives a smoother hazards rate curve with less variance, but the curve has some oscillatory behaviour. Thus, we need to use fewer knots, but also vary the placement of the knots. In the section around the exhaustion spike, we want more tightly placed knots that account for the timing of the exhaustion time. For the rest of the curve the knots must be spaced further apart to create a smooth curve. The knots chosen for the final spline specification are found by trying out different placements and number of knots until I am satisfied with how the curve follows the shape of the 0-degree specification, while being as smooth as possible.

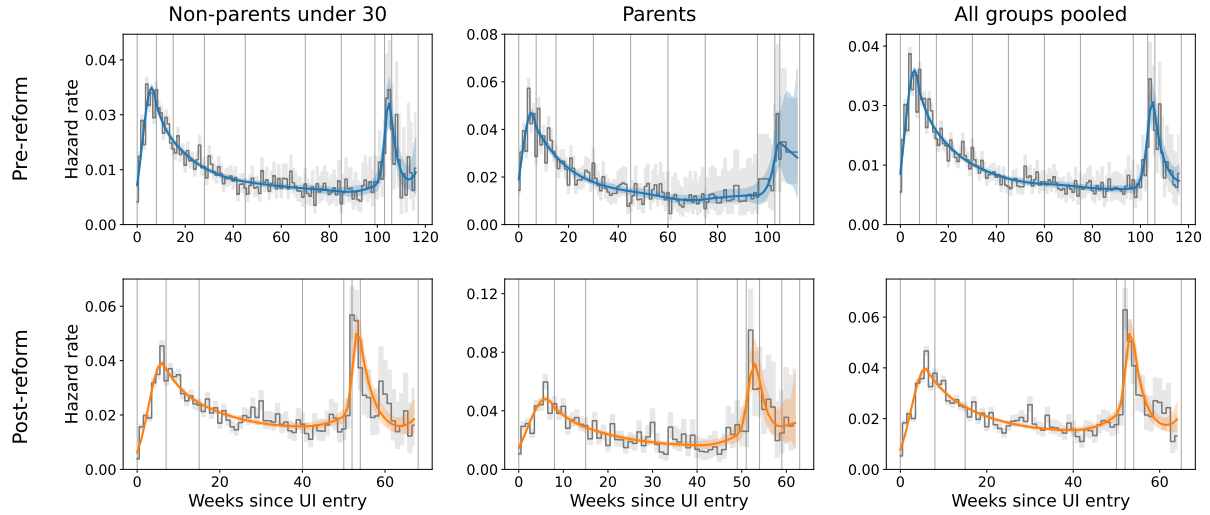
Figure 11: Empirical hazards under alternative B-spline specifications: non-parents 30+



Notes: Vertical lines mark the spline-knot locations in each specification. The degree-0 specification places a knot at every week.

This process is repeated for all groups as well as for the pooled sample of all three groups. In Figure 12 the final spline specifications is shown together with the 0-degree weekly hazards to show how they compare and how the knots are placed for each group. The group do not have identical knot placements, but are tweaked to have as representative a shape as possible. Though the placement around the exhaustion spike is kept largely the same.

Figure 12: Final spline specification compared to degree-0 weekly hazards



Notes: The grey step functions show the degree-0 weekly hazards, while the coloured curves show the corresponding final spline estimates. The vertical lines mark the knot locations used in the final spline specification.

Appendix D: Derivation of the Closed-Form Solution for Optimal Search Effort

The value function of the model is given by

$$V_t^U = \max_{s_t \in [0,1]} \{u(y_t) - \psi(s_t) + \delta [s_t V_{t+1}^E + (1 - s_t) V_{t+1}^U]\},$$

For the interior solution, optimal search effort satisfies the first-order condition

$$\frac{\partial V_t^U}{\partial s_t} = -\psi'(s_t) + \delta (V_{t+1}^E - V_{t+1}^U) = 0.$$

$$\Leftrightarrow \psi'(s_t^*) = \delta (V_{t+1}^E - V_{t+1}^U).$$

If ψ' is strictly increasing, it is invertible (Kristensen & Møller, 2021, Theorem 3.10), and the interior solution can be written as

$$s_t^* = (\psi')^{-1} (\delta [V_{t+1}^E - V_{t+1}^U]).$$

The derivative of the search-cost function used for the model is

$$\psi'(s_t) = \kappa s_t^\gamma.$$

Substituting this expression into the first-order condition gives the closed form solution

$$\kappa (s_t^*)^\gamma = \delta (V_{t+1}^E - V_{t+1}^U),$$

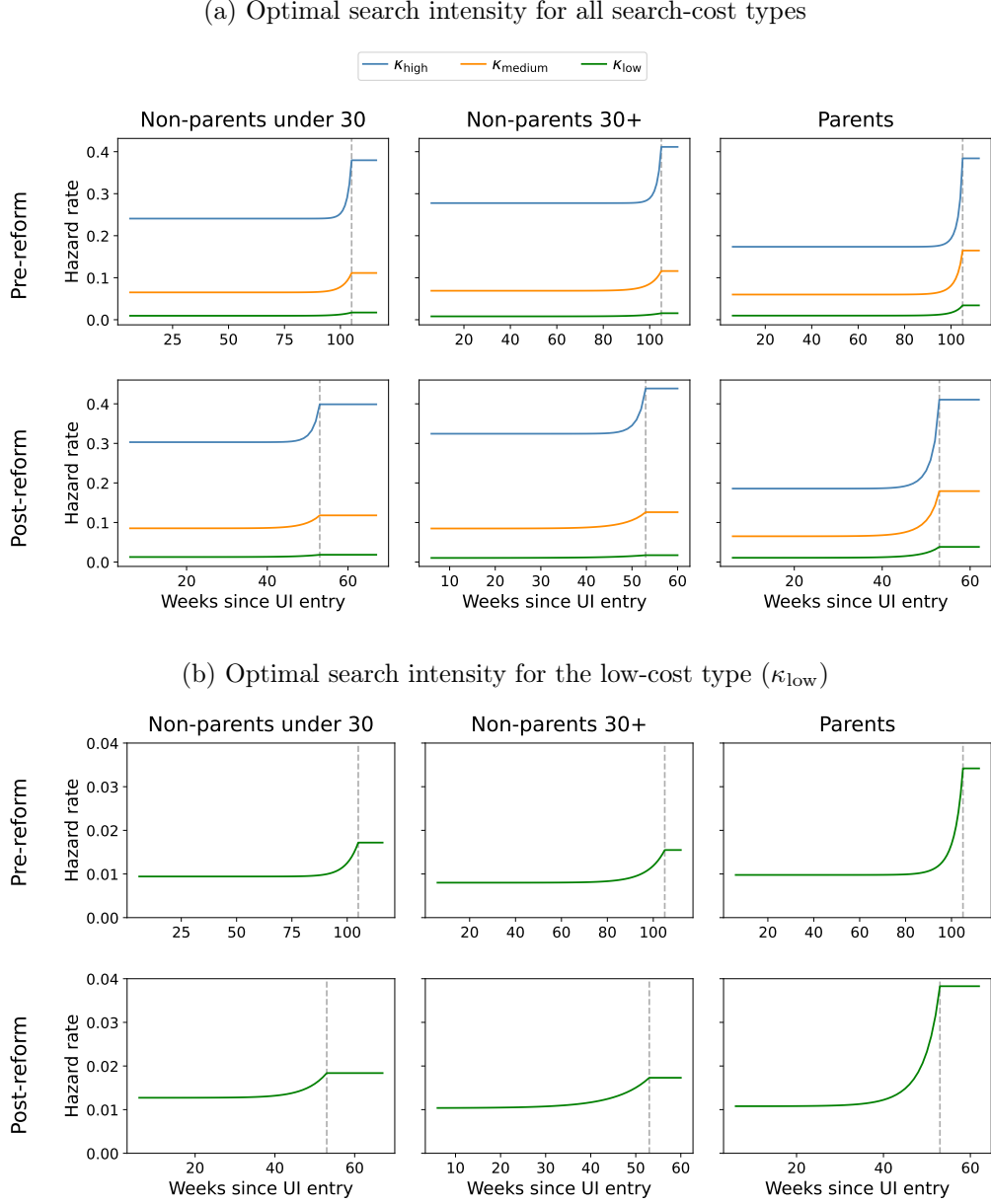
$$\Leftrightarrow s_t^* = \left[\frac{\delta}{\kappa} (V_{t+1}^E - V_{t+1}^U) \right]^{1/\gamma}.$$

This gives the interior solution for $s_t \in (0, 1)$. If the expression gives a value outside the feasible range $[0, 1]$, optimal search effort is set to the corresponding boundary, 0 or 1:

$$s_t^* = \min \left\{ 1, \max \left\{ 0, \left[\frac{\delta}{\kappa} (V_{t+1}^E - V_{t+1}^U) \right]^{1/\gamma} \right\} \right\}.$$

Appendix E: Optimal Search Intensity by Search-Cost Type

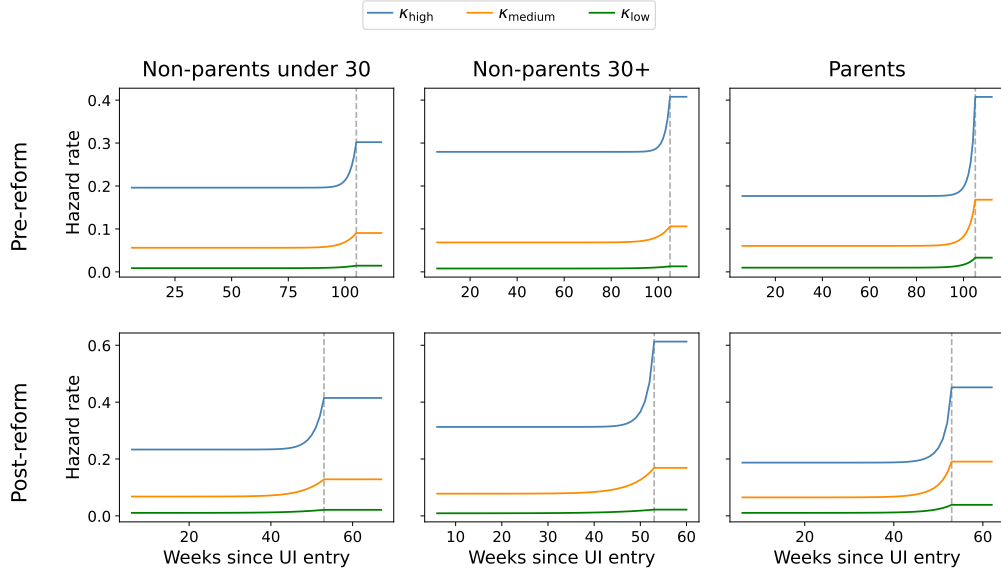
Figure 13: Optimal search intensity by search-cost type under log utility



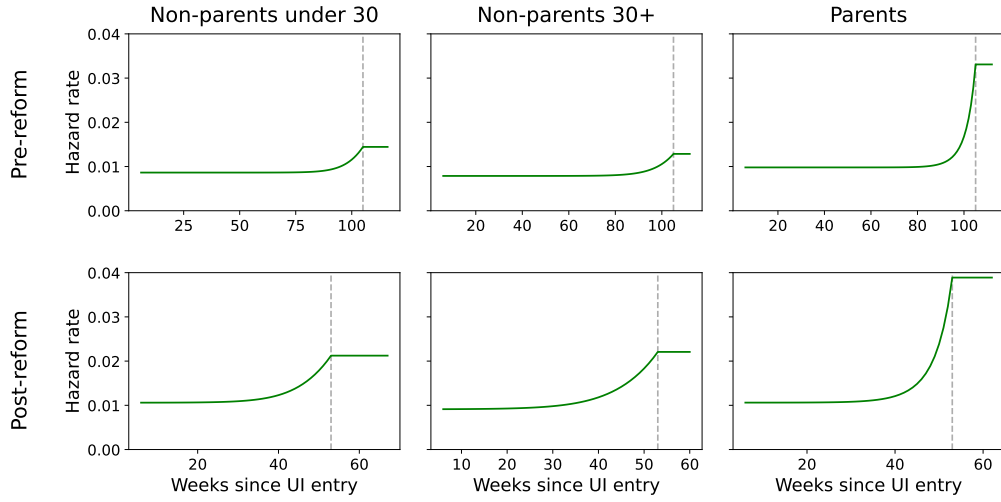
Notes: Panel (a) shows optimal search intensity for the high-, medium-, and low-search-cost types under the log utility specification. Panel (b) isolates the low-cost type because its search intensity is small relative to that of the other types. Dashed vertical lines mark UI exhaustion.

Figure 14: Optimal search intensity by search-cost type under shifted-log utility

(a) Optimal search intensity for all search-cost types



(b) Optimal search intensity for the low-cost type (κ_{low})



Notes: Panel (a) shows optimal search intensity for the high-, medium-, and low-search-cost types under the shifted-log utility specification. Panel (b) isolates the low-cost type because its search intensity is small relative to that of the other types. Dashed vertical lines mark UI exhaustion.